MIL-O-48202 (MU)
10 November 1972
SUPERSEDING
See 6.4

MILITARY SPECIFICATION

OGIVE AND OGIVE (ALTERNATIVE)

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for two types of ogive used on M424 and M440 projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surface for Organic Coating

MILITARY
MIL-P-116 - Preservation, Method of

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition

FSC 1110

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DRAWINGS

ORDNANCE CORPS
8786495 - Ogive and Ogive (Alternative)
8808362 - Packing and Marking for Storage and Shipment
8858976 - Carton, Packing, Ammunition, for Ogive, Projectile
EL-8786495 - Equipment List

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The ogive and alternative ogive shall comply with all requirements specified on Drawing (dwg.) 8786495, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 Tension test of ogive (alternative) (see dwg. 8786495). The ogive shall withstand a tension load of 8,000 ± 100 pounds applied slowly to the thread without any permanent visual or dimensional distortion or cracking of the ogive or weld when tested as specified in 4.4.1.

3.4 Paint adhesion. Paint adhesion shall be in accordance with paint adhesion, section 3, TT-C-490 when tested as specified in 4.4.2.

3.5 Workmanship.

3.5.1 Metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.3 Burrs. Parts shall be free from burrs.

3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserved the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19 - 20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22 - 23	20	41-43	28	87-97	36	520-1120	44
24 - 25	21	44-47	29	98-112	37	over 1120	45
26 - 27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Ogive, before painting (see 8786495 sheet 1 of 2).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Concentricity of forward thread and diameters, having a common axis, positioned one-half inch and 2.27 inches from the forward face with rear thread, including perpendicularity of threaded surfaces to right and left faces.....Gage
102. Concentricity of right external surface with left thread, including perpendicularity to left face.....Gage
103. Concentricity of hole in internal neck with left thread including perpendicularity to left face .....Gage
104. Pitch diameter of right thread .....Gage
105. Minor diameter of right thread.....Gage
106. Pitch diameter of left thread.....Gage
107. Minor diameter of left thread.....Gage
108. Wall thickness, min .....Gage
109. Diameter of hole in internal neck.....Gage
110. Total length.....Gage
111. Internal diameter of flat surface right of left thread.Gage
112. Distance between wrench slots.....Gage
113. Width of wrench slots, min.....Gage

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114. Location of wrench slots.....Gage
115. Length of right thread, min.....Gage
116. Thickness of internal tapered wall, min.....Gage
117. Length of hole in internal neck, min.....Gage
118. Length from left face to left datum diameter.....Gage
119. Length from left face to right datum diameter.....Gage
120. Length from left face to left face of internal hole.....Gage
121. Length of left thread, min.....Gage
122. Thickness at end of internal wall, min.....Gage
123. Material (see 3.1).....Visual

Minor:

201. No radius improper or missing.....Visual
202. Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8786495....Visual
203. Workmanship (see 3.5).....Visual

4.3.2.1.2 Ogive (see dwg. 8786495, sheet 1 of 2).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:

201. No paint improper, including no bare spots.....Visual
202. Workmanship (see 3.5).....Visual

4.3.2.1.3 Ogive, before painting (alternative), (see dwg. 8786495 sheet 2 of 2).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Concentricity of forward thread and diameters, having a common axis, positioned one-half inch and 2.27 inches from the right face with left thread, including perpendicularity of threaded surfaces to right and left faces.....Gage
102. Concentricity of forward external surface with left thread, including perpendicularity to left face.....Gage
103. Concentricity of hole in partition wall with left thread, including perpendicularity to left face.....Gage
104. Pitch diameter of right thread.....Gage

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105. Minor diameter of right thread.....Gage
106. Pitch diameter of left thread .....Gage
107. Minor diameter of left thread.....Gage
108. Wall thickness, min.....Gage
109. Diameter of hole in partition wall .....Gage
110. Total length.....Gage
111. Length from left face to right datum diameter.....Gage
112. Length from left face to left datum diameter.....Gage
113. Internal diameter of flat surface right of left thread.....Gage
114. Distance between wrench slots.....Gage
115. Width of wrench slots, min.....Gage
116. Location of wrench slots .....Gage
117. Length of right thread, min.....Gage
118. Length of left thread, min.....Gage
119. Thickness of partition wall .....Gage
120. Length from right face of partition wall to left face.....Gage
121. Material (see 3.1).....Visual

Minor:

201. No radius improper or missing .....Visual
202. No improper weld .....Visual
203. Objective evidence (see 6.2)available for protective finish in accordance with dwg. 8786495 .Visual
204. Workmanship (see 3.5).....Visual

4.3.2.1.4 Ogive (alternate) (see dwg. 8786495, sheet 2 of 2).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:

201. No paint improper including no bare spots.....Visual
202. Workmanship (see 3.7).....Visual

4.3.2.1.5 Carton, Packing, Ammunition, for Ogive, Projectile (see dwg. 8858976 and section 5).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:

101.	Package material (see 3.1)	Visual
102.	Carton length, width and height before packing, min	*Measure
103.	Bag length and width before packing, min	*Measure
104.	Ogive packed tight	Manual
105.	Carton sealed properly	Visual
106.	Corners of carton blunted	Visual
107.	Air exhausted from bag	Manual
108.	Proper heat seal	Visual
109.	Objective evidence (see 6.2) available for heat seal test in accordance with dwg. 8858976	Visual

*Measure with suitable scale.

Minor:

201.	Marking	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.5 Packing and Marking for Storage and Shipments (see dwg. 8808362 and section 5).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Gross load weight, max	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Package material (see 3.1)	Visual

*Determine weight with a suitable scale.

Minor:

201.	Marking	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in order specified.

Test	Test classification
a. Tension	Critical
b. Paint adhesion	Minor

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None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Tension test of ogive (alternative) (see 3.3). One sample shall be tested as specified in 4.4.1 for each production lot of 110 or less and one additional sample for every fourty (40) parts produced over lot size of 110. If the sample fails to comply with requirement 3.3, the production lot of ogives (alternative) shall be rejected.

4.3.3.2 Paint adhesion (see 3.4). Each part or assembly shall be tested as specified in 4.4.2. Any part or assembly which fails to comply with 3.4 shall be classified defective and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8786495). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Tension test of ogive (alternative) (see 3.3 and dwg. 8786495). The ogive assembly shall have a load applied by an approved testing device, through a suitable adapter screwed into the tube thread. These adapters shall rely only on thread engagement to exert the tension force on the ogive. The load shall be applied in a minimum time of 10 seconds and permitted to dwell at full load for a minimum of 10 seconds.

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4.4.2 Paint adhesion (see 3.4 and dwg. 8786495). Paint adhesion shall be tested in accordance with paint adhesion, section 4, Federal Specification TT-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The ogive shall be packaged in accordance with dwgs. 8808362 and 8858976.

5.1.2 Level C. The ogive shall be packaged in accordance with dwgs. 8808362 and 8858976 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The ogive shall be packed in accordance with dwgs. 8808362 and 8858976.

5.3 Marking. Marking shall be as specified on dwgs. 8808362 and 8858976.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1520 dated 29 August 1969.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1110-A034

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DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-O-48202(MU) Ogive and Ogive (Alternative)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 JAN 64 WHICH MAY BE USED.

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